

Simulation of Flexible Coplanar Waveguide

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1 Introduction

Coplanar Waveguides (CPWs) are among the most widely used types of transmission lines due to their straightforward planar geometry and ease of integration with microwave circuits. A typical CPW consists of a central signal conductor printed on a dielectric substrate and flanked by two ground conductors, separated from the signal line by narrow air gaps. In standard applications, such as signal transmission in integrated circuits or telecommunications, the width of the signal line and the gaps are typically on the order of tens to hundreds of micrometers[6], while the substrate thickness generally ranges from $100\text{ }\mu\text{m}$ to few millimeters, depending on the dielectric material employed.

At Fondazione Bruno Kessler (FBK), we are currently developing flexible transmission lines composed of $25\text{ }\mu\text{m}$ thick Kapton dielectric layers and patterned aluminum traces with a width of $90\text{ }\mu\text{m}$ and a thickness of $20\text{ }\mu\text{m}$. These structures, while complex in their physical realization, can in a first-order approximation be modeled as CPW. Figure 1 shows a schematic representation of the FBK flexible CPW, including all relevant dimensional parameters.

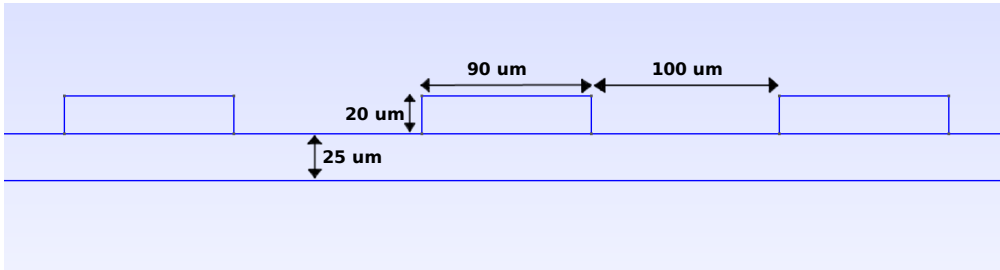


Figure 1: Schematic view of the aluminum-kapton wave guide.

As shown in the schematic view, the ground traces and signal traces dimensions are comparable: this configuration takes the name of Finite Ground Coplanar Waveguide (FGCPW). FGCPW equipped with the particular material choice of FBK haven't been studied extensively yet, so it will be important to characterize them with both real measures and simulations.

A widely used method to characterize transmission lines is through the analysis of scattering parameters (S-parameters)[1]. For a two-port network, these parameters can be expressed as a 2×2 matrix. The diagonal terms, S_{11} and S_{22} , represent the reflection coefficients at each port, and are directly related to impedance mismatches between the line's characteristic impedance Z_0 and the source (Z_S) and load (Z_L) impedances. The off-diagonal terms, S_{21} and S_{12} , represent the forward and reverse transmission coefficients, and are associated with insertion loss, which depends on both the material losses and the physical length of the line. The insertion losses are defined as [8]:

$$\text{Insertion Losses} = \frac{-20 \log_{10}(|S_{21}|)}{L} \quad (1)$$

Due to the structural and material differences between standard CPWs and our flexible FGCPW, it is essential to analyze its behavior across a broad frequency range. While physical measurements using a Vector Network Analyzer (VNA) are ultimately required, numerical simulations provide a valuable preliminary tool for characterizing S-parameters.

2 Flex FGCPW and simulation procedure

2.1 Transmission line structure

The FGCPW under analysis consists of a $25 \mu m$ thick Kapton dielectric layer and copper signal/ground traces with a thickness of $20 \mu m$. Copper was selected instead of aluminum for these simulations to enable direct comparison with predictions from the Sierra PCB toolkit, which assumes copper by default, but a final simulation with aluminum has been performed.

Simulations were performed with copper conductivities ranging from 58.0 to $59.6 MS/m$ (corresponding to resistivities between 1.68×10^{-8} and $1.72 \times 10^{-8} \Omega \cdot m$), as well as an additional value of $62.5 MS/m$ ($1.60 \times 10^{-8} \Omega m$) to account for ideal copper. Kapton exhibits a low dielectric loss, with a dissipation factor $\tan \delta$ of 0.0013 at $10 GHz$, significantly lower than the value of other common substrates (e.g. $22 \mu m$ Rogers4350B with $\tan \delta = 0.0029$ at $10 GHz$), indicating minimal dielectric energy loss.

The signal lines are $90 \mu m$ wide, with $100 \mu m$ gaps between the signal and ground conductors. The total length of the transmission line is $1 mm$. Based on these parameters, the Sierra PCB toolkit returns a value for the characteristic impedance Z_0 of 122.64Ω , while other tools such as TXLINE[7] or ATCL2[5] estimate lower values between 105 and 110Ω . No surface finish or encapsulation is applied to the copper traces, which are exposed to air on the top side.

2.2 Simulation

Simulations were conducted using the AWS Palace Toolkit[3], employing the lumped port excitation method. This setup allows both ends of the signal trace to be terminated with specified Z_S and Z_L impedances, while an excitation signal is injected via

the ports.

The ports are positioned parallel to the substrate surface and occupy the $100 \mu m$ air gap near the trace ends. Key parameters affecting the simulation accuracy include impedance matching and port size. Impedance mismatches primarily influence S_{11} and S_{22} , while excessive port lengths may shorten the effective transmission line, impacting S_{21} and S_{12} . Given the symmetry of the S-matrix for a reciprocal, linear network, we refer only to S_{11} and S_{21} in subsequent discussions.

Another crucial point of the analysis is the estimation of the total insertion loss since, as shown in Equation 1, this parameter provides a further prediction on the value of S_{21} . The total insertion loss is the sum of two contributes: dielectric losses α_D due to the polarization of the substrate and conductor losses α_C due to Ohmic losses[2]. Following the IPC-2251 standard[2, 4], these terms can be calculated as follows:

$$\alpha_D = 8.686\pi f \sqrt{\mu_0 \varepsilon_0 \varepsilon_r} \tan \delta \text{ dB/m} \quad (2)$$

$$\alpha_C = \frac{8.686 R_s \sqrt{\varepsilon_r^{eff}}}{480\pi K(k_s) K'(k_s) (1 - k_s^2)} \left(\frac{1}{a} \left[\pi + \log \left(\frac{8\pi a(1 - k_s)}{t(1 + k_s)} \right) \right] + \frac{1}{b} \left[\pi + \log \left(\frac{8\pi b(1 - k_s)}{t(1 + k_s)} \right) \right] \right) \text{ dB/m} \quad (3)$$

where f is the frequency, R_s is the surface resistance, $a = t/2$, $b = a + g$, $k_s = a/b$ and K, K' are complete elliptic integrals of the first kind, referring to t, w, g as trace thickness, width and gap between traces respectively. In Section 3, we will show the results of α_D , α_C , using them as insertion loss prediction and labeling them as IPC in the plot legends.

An important aspect for this analysis is that both α_D and α_C are functions of the frequency. This is trivial from Equation 2, while for α_C the relation is inside R_s . Indeed, the following relations hold:

$$R_s = \frac{1}{\sigma t} \quad \delta \gg t \quad (4)$$

$$R_s = \frac{1}{\sigma \delta} \quad \delta \ll t \quad (5)$$

where σ and t are the conductivity and the thickness of the metal trace, and δ is the skin depth, defined as

$$\delta = \frac{1}{\sqrt{\pi \mu \sigma f}} \quad (6)$$

Equation 4 shows that R_s is constant when skin depth is higher than the typical conductor dimensions while increases with the frequency due to skin effect (Equation 5).

Furthermore, conductor losses trend can be useful to evaluate when the skin effect become relevant in the frequency domain. We can choose an arbitrary factor $N > 1$ and assume that skin effect is dominant when

$$\alpha_C^{HF} - \alpha_C^{LF} \geq N \alpha_C^{LF} \quad (7)$$

or equivalently when $\delta \leq t/N$. Solving these equations for the frequency we obtain

$$f_{thr} \geq (N \alpha_C^{LF})^2 \left(\frac{2Z_0 \sigma w}{8.686} \right)^2 \frac{1}{\pi \mu \sigma} \quad (8)$$

Using Equations 2,3 it is possible to compute the total insertion losses which can be used as a reference for the evaluation of the simulation result.

Finally, we defined a quantitative χ^2 -based method to assess the optimal parameter space that is, the combination of $Z_{S,L}$, mesh size, and conductivity, selected for each simulation. For each case, we assumed a specific value for the data dispersion around a chosen reference (either from Sierra Toolkit, IPC insertion losses predictions or VNA measurements) and calculated the corresponding χ^2 . When $\chi^2/\text{dof} \approx 1$, we can conclude that the simulation exhibits an average deviation consistent with the assumed dispersion and that the selected parameter space provides the best agreement with the reference data.

3 Results

Initial simulations were performed to identify the source and load impedance values that best match the characteristic impedance Z_0 . When Z_S and Z_L are close to Z_0 , the reflected signal is minimized, resulting in the highest magnitude curve for S_{11} . Figures 2-3 show the frequency dependence of S_{11} and S_{21} for different impedance values.

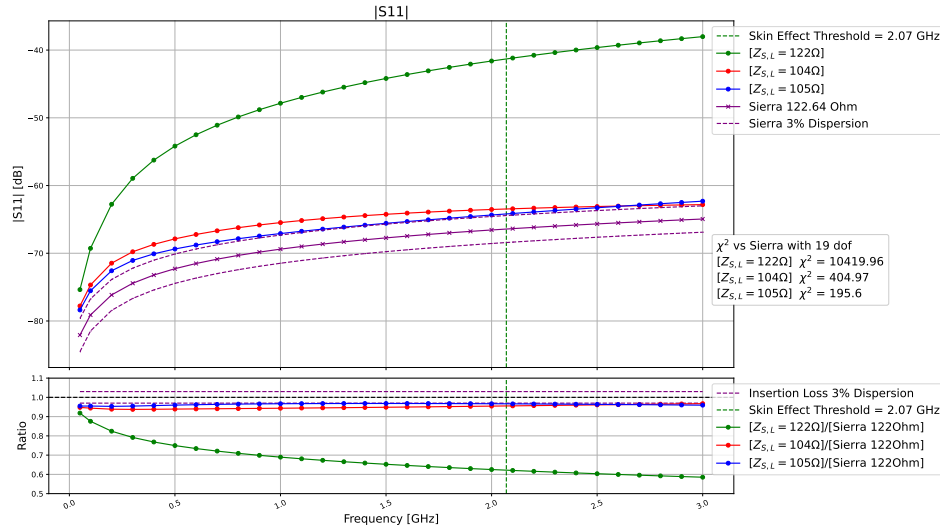


Figure 2: S_{11} plot for different port impedance values. f_{thr} is set to 2.06 GHz and χ^2 values are shown assuming Sierra as reference and 3% dispersion (dashed yellow line).

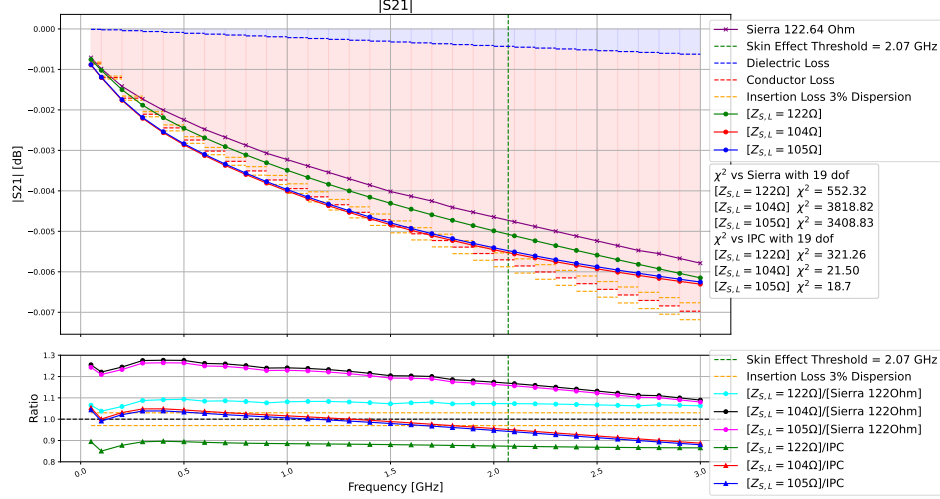


Figure 3: S_{21} plot for different port impedance values. χ^2 are calculated using as reference both Sierra (purple line) and IPC predicted insertion losses (dash red lines). Ratios are evaluated comparing data with IPC and (green, red, blue lines) and with Sierra (cyan, black, pink lines).

Figure 2 show the reflected signal attenuation under different impedance hypothesis. The vertical dashed green line represents the skin effect threshold at 2.06 GHz calculated with Equation 8 using $N = 14$. The purple line predicted by Sierra was calculated under the 122.64 Ω matching impedance hypothesis, but the simulation that best aligns with this reference used an impedance of 105 Ω . Figure 3 shows the simulated $Z_{S,L}$ -dependent insertion losses and the Sierra predicted trend. The blue and red dashed lines are the IPC predicted dielectric and conductor losses respectively and they were used as a second reference for our S_{21} values, while the yellow dashed lines represent a 3% dispersion around the IPC insertion losses. The S_{21} results are well aligned to the insertion losses calculated with Equations 2-3 considering a Z_0 value of 105 Ω , while both Sierra data and simulation at $Z_0 = 122.64 \Omega$ are the most distant points from the predicted trend. This is also confirmed from the χ^2 study with 19 dof, which shows values of 321.26, 21.50 and 18.77 for the $Z_{S,L} = 122.64, 104, 105\Omega$ respectively assuming a 3% dispersion from IPC predicted insertion losses. Much worse χ^2 are exhibit if we assume Sierra curve as reference ($\chi^2_{122\Omega} = 552.32$, $\chi^2_{104\Omega} = 3818.82$, $\chi^2_{105\Omega} = 3408.83$). For the previous and following plots, χ^2 values are calculated excluding points with $f \geq f_{thr}$ since, when the skin effect becomes dominant, we would require a more complex mesh refined along the conductor surfaces with approximately the size of skin depth ($\delta \approx 1 \mu m$ at 2 GHz). We conclude that Sierra PCB Toolkit is a good starting point for this analysis but, due to our particular choices of FGCPW geometry and materials instead of classical CPW, it cannot predict the correct results for our case.

For all subsequent simulations, a matched impedance of 105 Ω is adopted and we

will not show anymore the Sierra reference. The previous plots were generated with a minimum mesh element size of $22.5 \mu m$ and copper conductivity of $59.6 MS/m$. In the next part of the analysis, these parameters were varied to study their influence on simulation accuracy and to better match the IPC reference. Results are shown in the following plots.

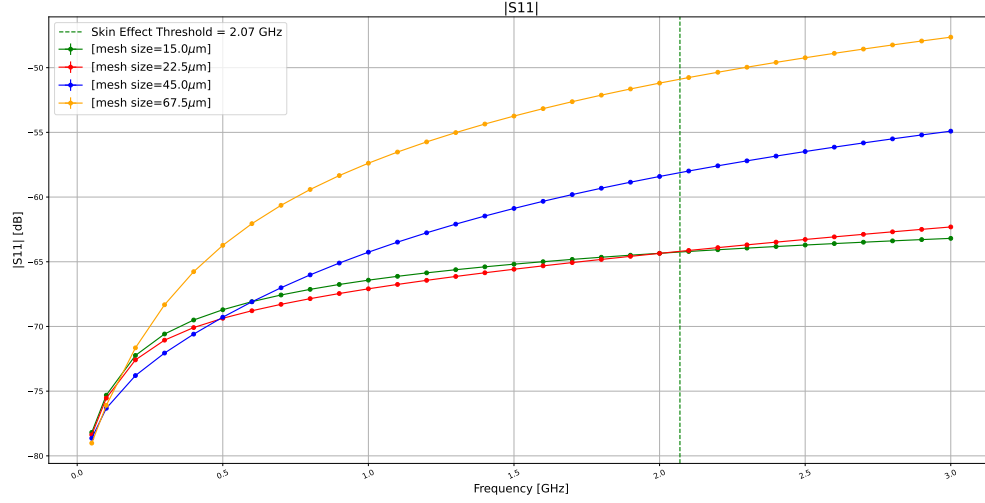


Figure 4: S_{11} compared results for different mesh sizes and $\sigma = 59.6 MS/m$.

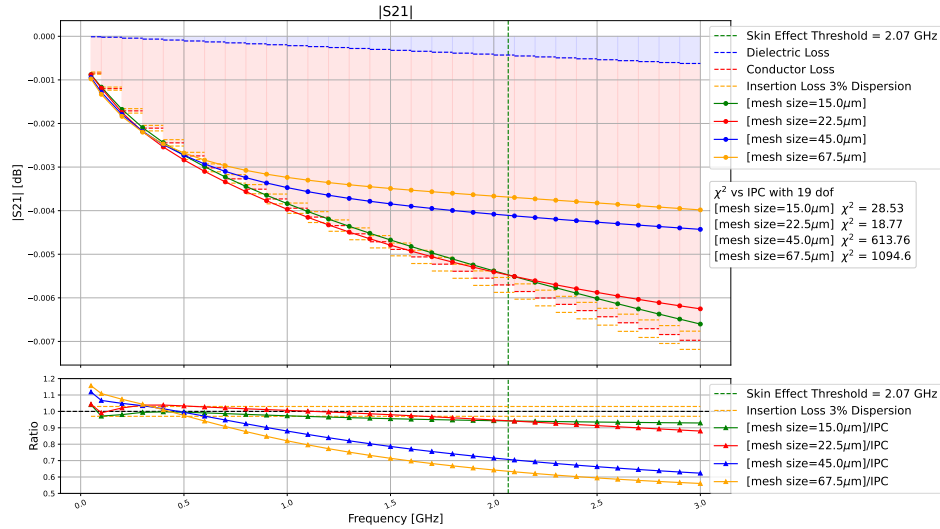


Figure 5: S_{21} compared results for different mesh sizes and $\sigma = 59.6 MS/m$.

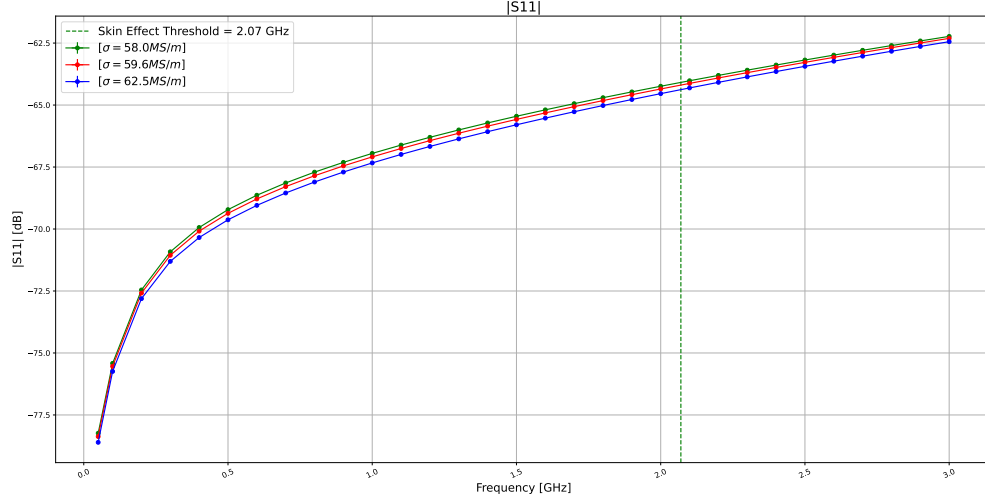


Figure 6: S_{11} compared results for different copper conductivity and mesh size = $22.5 \mu m$.

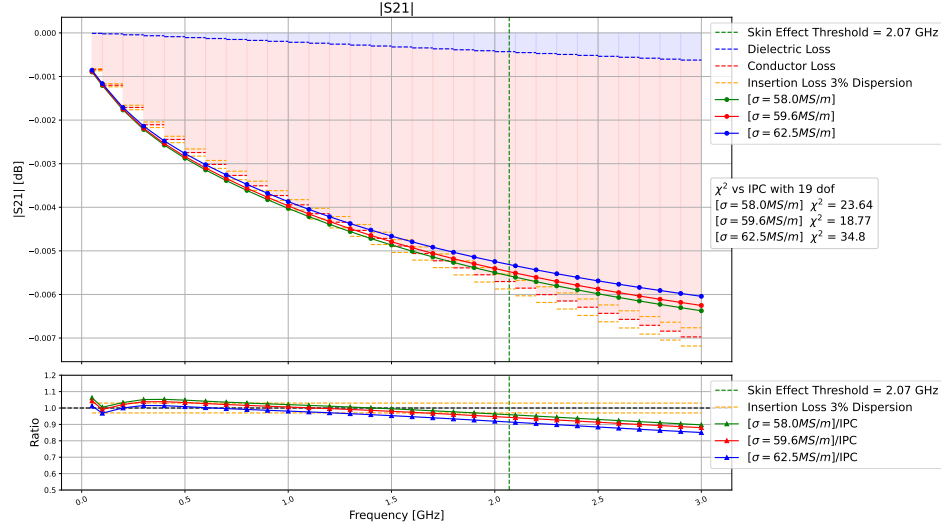


Figure 7: S_{21} compared results for different copper conductivity and mesh size = $22.5 \mu m$.

Figures 4-5 illustrate that using coarse mesh sizes (45 or $90 \mu m$) leads to S_{11} deviations (w.r.t. finer mesh sizes) of more than 10 dB and deviations of approximately 40% of the predicted insertion loss value at high frequencies. When the mesh size becomes

comparable to the conductor thickness, the results converge and further refinement yields negligible improvements below the skin effect threshold. In fact, for frequencies above the threshold, the $15\ \mu\text{m}$ simulation seems better aligned with the IPC predictions, as confirmed by χ^2 analysis. Consequently, a mesh size of $22.5\ \mu\text{m}$ (one-fourth of the trace width) is selected, but at higher frequency will be necessary to use finer meshes.

Figures 6–7 show the effect of varying the copper conductivity. The green and red curves correspond to conductivities of 58.0 and 59.6 MS/m, respectively, while the blue curve represents an ideal copper scenario with a very high conductivity of 62.5 MS/m, corresponding to a low resistivity of $1.60 \times 10^{-8}\ \Omega\cdot\text{m}$. As expected from Equation 3, the insertion loss and S_{21} are directly affected by the conductivity, with lower conductivity leading to higher power losses, whereas S_{11} is essentially independent of this parameter. In Figure 7, the reference insertion losses are computed using $\sigma = 59.6\ \text{MS/m}$; it is therefore natural that the best χ^2 value is obtained when simulating with this same conductivity. Nevertheless, this result provides an additional indication of the robustness of the previously selected parameter set.

Finally, having established the optimized simulation parameters (a mesh size of $22.5\ \mu\text{m}$ and $Z_{S,L} = 105\ \Omega$), we can return to the actual FBK coplanar waveguide configuration, in which the conductors are made of aluminum with a conductivity of $\sigma = 3.31 \times 10^7\ \text{S/m}$. The resulting simulations are then compared with both the VNA measurements and the IPC insertion losses predictions, assuming an average relative error of 3% over the entire frequency range, as determined from the χ^2 analysis.

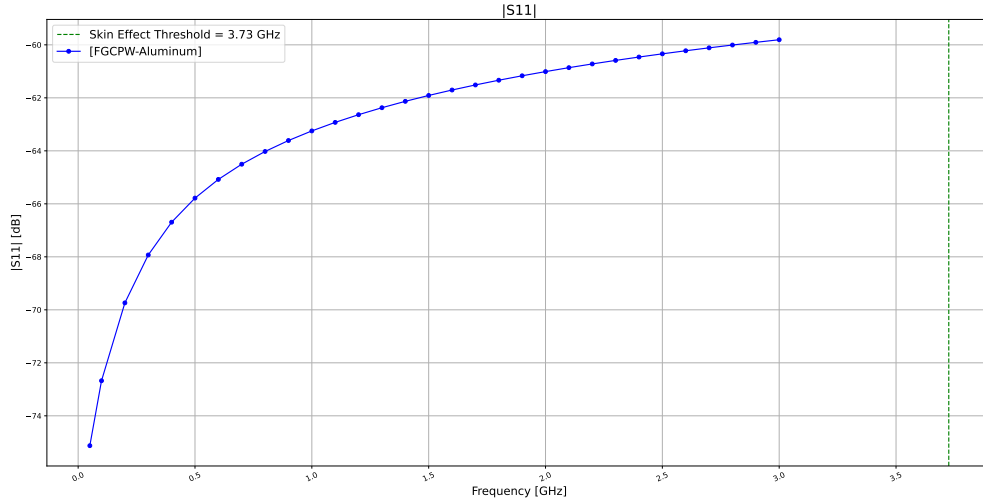


Figure 8: Reflection losses simulation for aluminum-CPW.

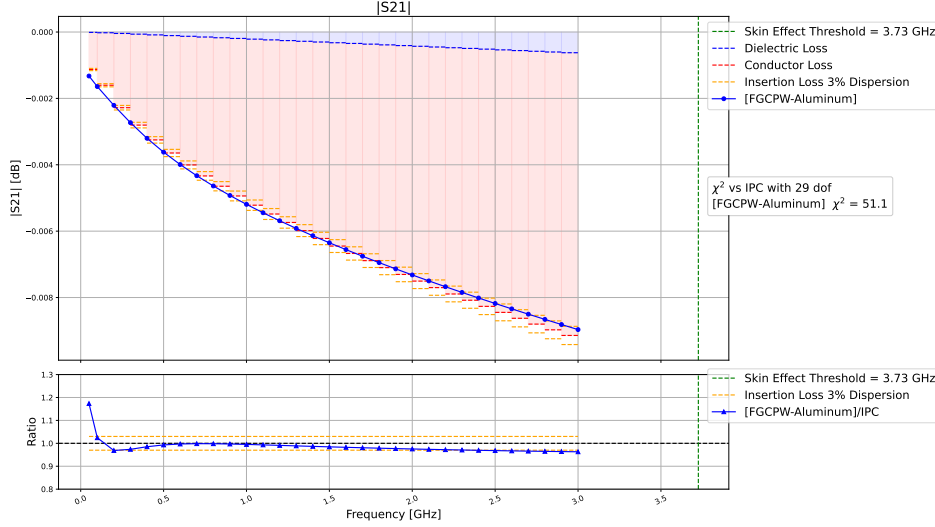


Figure 9: Insertion losses simulation for aluminum-CPW.

4 Conclusion

In this work, we studied the dependence of FGCPW simulations on several parameters, such as mesh refinement, conductor properties, and matching impedance. We calibrated these parameters by calculating the S-matrix and identifying the set of values that minimize both reflection and insertion losses of the transmission line, using a χ^2 -based method to assess the optimal parameter space for simulating the current Device Under Test. This method has been tested on our CPW, which represents a relatively simple case, but it can be applied to any other geometry. Finally, after comparing the simulation results with VNA measurements and finding a high level of agreement, we can now rely on the simulation and extract additional parameters such as the propagation constant, phase delay, and the R , L , C , G values of the transmission line.

Furthermore, using the χ^2 analysis, we can claim Palace simulation reproduce the reference within an error of about 3%, but this value can probably be improved reducing mesh sizes at values close to the skin depth δ at high frequencies ($\delta \approx 2 \mu m$ at 1 GHz for both copper and aluminum).

References

- [1] W.R. Eisenstadt and Y. Eo. “S-parameter-based IC interconnect transmission line characterization”. In: *IEEE Transactions on Components, Hybrids, and Manufacturing Technology* 15.4 (1992), pp. 483–490. DOI: 10.1109/33.159877.

- [2] G. Ghione and C.U. Naldi. “A new analytical, CAD-oriented model for the ohmic and radiation losses of asymmetric coplanar waveguides in hybrid and monolithic MIC’s”. In: *Gallium Arsenide Applications Symposium. GAAS 1992, 27-29, Noordwijk, The Netherlands* (1992).
- [3] S. Grimberg, H. Carson, and A. Keller. URL: <https://awslabs.github.io/palace/stable/>.
- [4] IPC-2251 Task Group. “Design Guide for the Packaging of High Speed Electronics Circuits”. In: <https://www.electronics.org/ipc-standards> (2003).
- [5] D. Kirkby. URL: <http://www.hdtvprimer.com/kq6qv/atlc2.html>.
- [6] B. Nataraj and K.R. Prabha. “Analysis and Modeling of various Tapered Coplanar Waveguide”. In: *IOP Conf. Ser.: Mater. Sci. Eng.* 1084 (2021), p. 012044.
- [7] R.C. Pastorici. URL: <https://www.qsl.net/yo4hfu/Microstrip.html>.
- [8] X. Wu et al. “Surface finish effects on high-speed signal degradation”. In: *IEEE Trans. Adv. Packag.* 30 (2008), pp. 182–189.